

Construction of Mimetic Curl Operators Using the MOLE Gradient

1 Preliminaries

The Mimetic Operators Library Enhanced (MOLE) provides a one-dimensional mimetic gradient operator

$$\mathbf{G}_x = \mathbf{grad}(k, m, \Delta x) \in \mathbb{R}^{(m+1) \times (m+2)},$$

where k is the order of accuracy and m is the number of cells. This operator maps scalar values at *cell centers plus boundary points* ($m + 2$ positions) to derivatives at *cell nodes* ($m + 1$ positions). The staggered positions are:

- **Nodes** (cell boundaries): $x_0, x_1, \dots, x_m$ ($m + 1$ points).
- **Centers + boundary**: $x_0, x_{1/2}, x_{3/2}, \dots, x_{m-1/2}, x_m$ ($m + 2$ points).

All multi-dimensional arrays use **ndgrid ordering** (column-major with x varying fastest), consistent with MOLE's convention. The Kronecker product $\otimes$ is used to extend 1D operators to higher dimensions.

2 2D Curl Operator

2.1 Continuous definition

For a 2D vector field $\mathbf{F} = (F_x, F_y)$, the scalar curl is

$$(\nabla \times \mathbf{F})_z = \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}.$$

2.2 Staggering

On a grid with m cells in x and n cells in y :

Component	Grid positions	Size
F_x (x-faces)	x : nodes ($m + 1$), y : centers+bnd ($n + 2$)	$(m + 1)(n + 2)$
F_y (y-faces)	x : centers+bnd ($m + 2$), y : nodes ($n + 1$)	$(m + 2)(n + 1)$
$\text{curl}(\mathbf{F})$ (nodes)	x : nodes ($m + 1$), y : nodes ($n + 1$)	$(m + 1)(n + 1)$

The input vector is assembled as $\mathbf{F} = [F_x; F_y]$ with total length $(m + 1)(n + 2) + (m + 2)(n + 1)$.

2.3 Construction

Let $\mathbf{G}_x = \text{grad}(k, m, \Delta x) \in \mathbb{R}^{(m+1) \times (m+2)}$ and $\mathbf{G}_y = \text{grad}(k, n, \Delta y) \in \mathbb{R}^{(n+1) \times (n+2)}$. Define the identity matrices $\mathbf{I}_{m+1} \in \mathbb{R}^{(m+1) \times (m+1)}$ and $\mathbf{I}_{n+1} \in \mathbb{R}^{(n+1) \times (n+1)}$.

Term $\partial F_x / \partial y$. F_x lives on an $(m+1) \times (n+2)$ grid. Applying $\mathbf{G}_y$ along y while preserving the x dimension gives

$$\frac{\partial F_x}{\partial y} \longleftrightarrow (\mathbf{G}_y \otimes \mathbf{I}_{m+1}) F_x, \quad (\mathbf{G}_y \otimes \mathbf{I}_{m+1}) \in \mathbb{R}^{(n+1)(m+1) \times (n+2)(m+1)}.$$

Term $\partial F_y / \partial x$. F_y lives on an $(m+2) \times (n+1)$ grid. Applying $\mathbf{G}_x$ along x while preserving the y dimension gives

$$\frac{\partial F_y}{\partial x} \longleftrightarrow (\mathbf{I}_{n+1} \otimes \mathbf{G}_x) F_y, \quad (\mathbf{I}_{n+1} \otimes \mathbf{G}_x) \in \mathbb{R}^{(n+1)(m+1) \times (n+1)(m+2)}.$$

Assembled operator. The 2D curl matrix acts on $\mathbf{F} = [F_x; F_y]$ as:

$$\boxed{\mathbf{C}_{2D} = \begin{bmatrix} -(\mathbf{G}_y \otimes \mathbf{I}_{m+1}) & (\mathbf{I}_{n+1} \otimes \mathbf{G}_x) \end{bmatrix} \in \mathbb{R}^{(m+1)(n+1) \times [(m+1)(n+2) + (m+2)(n+1)]}}$$

so that $\mathbf{C}_{2D} \mathbf{F} = \partial F_y / \partial x - \partial F_x / \partial y$.

2.4 Curl of gradient identity

The full 2D gradient (without boundary injection) maps a scalar field ϕ at $(m+2)(n+2)$ cell-center positions to a face-staggered vector:

$$\mathbf{G}_{\text{full}} = \begin{bmatrix} \mathbf{I}_{n+2} \otimes \mathbf{G}_x \\ \mathbf{G}_y \otimes \mathbf{I}_{m+2} \end{bmatrix}.$$

Then

$$\begin{aligned} \mathbf{C}_{2D} \mathbf{G}_{\text{full}} &= -(\mathbf{G}_y \otimes \mathbf{I}_{m+1})(\mathbf{I}_{n+2} \otimes \mathbf{G}_x) + (\mathbf{I}_{n+1} \otimes \mathbf{G}_x)(\mathbf{G}_y \otimes \mathbf{I}_{m+2}) \\ &= -(\mathbf{G}_y \mathbf{I}_{n+2}) \otimes (\mathbf{I}_{m+1} \mathbf{G}_x) + (\mathbf{I}_{n+1} \mathbf{G}_y) \otimes (\mathbf{G}_x \mathbf{I}_{m+2}) \\ &= -\mathbf{G}_y \otimes \mathbf{G}_x + \mathbf{G}_y \otimes \mathbf{G}_x \\ &= \mathbf{0}, \end{aligned}$$

using the mixed-product property $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD})$. This confirms $\nabla \times (\nabla \phi) = 0$ holds exactly at the discrete level.

3 3D Curl Operator

3.1 Continuous definition

For a 3D vector field $\mathbf{F} = (F_x, F_y, F_z)$:

$$\nabla \times \mathbf{F} = \begin{pmatrix} \partial F_z / \partial y - \partial F_y / \partial z \\ \partial F_x / \partial z - \partial F_z / \partial x \\ \partial F_y / \partial x - \partial F_x / \partial y \end{pmatrix}.$$

3.2 Staggering

On a grid with $m \times n \times o$ cells:

	Grid dimensions	Size
<i>Input: face-staggered field</i>		
F_x (x-faces)	$(m+1) \times (n+2) \times (o+2)$	$(m+1)(n+2)(o+2)$
F_y (y-faces)	$(m+2) \times (n+1) \times (o+2)$	$(m+2)(n+1)(o+2)$
F_z (z-faces)	$(m+2) \times (n+2) \times (o+1)$	$(m+2)(n+2)(o+1)$
<i>Output: edge-staggered curl</i>		
C_x (x-edges)	$(m+2) \times (n+1) \times (o+1)$	$(m+2)(n+1)(o+1)$
C_y (y-edges)	$(m+1) \times (n+2) \times (o+1)$	$(m+1)(n+2)(o+1)$
C_z (z-edges)	$(m+1) \times (n+1) \times (o+2)$	$(m+1)(n+1)(o+2)$

3.3 Construction

Let $\mathbf{G}_x$, $\mathbf{G}_y$, $\mathbf{G}_z$ be the 1D mimetic gradients of sizes $(m+1) \times (m+2)$, $(n+1) \times (n+2)$, and $(o+1) \times (o+2)$, respectively. Define identity matrices $\mathbf{I}_p$ of appropriate sizes with subscripts indicating dimension.

Each partial derivative is extended to 3D via Kronecker products. For a field on grid $(d_x \times d_y \times d_z)$, the operator applying $\mathbf{G}_\alpha$ along dimension α is:

$$\frac{\partial}{\partial z} \longleftrightarrow \mathbf{G}_z \otimes \mathbf{I}_{d_y} \otimes \mathbf{I}_{d_x}, \quad \frac{\partial}{\partial y} \longleftrightarrow \mathbf{I}_{d_z} \otimes \mathbf{G}_y \otimes \mathbf{I}_{d_x}, \quad \frac{\partial}{\partial x} \longleftrightarrow \mathbf{I}_{d_z} \otimes \mathbf{I}_{d_y} \otimes \mathbf{G}_x.$$

Applying this rule to each term of the curl, with the appropriate grid dimensions for each input component:

Term	Input grid	Kronecker form
$\partial F_z / \partial y$	$(m+2, n+2, o+1)$	$\mathbf{I}_{o+1} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+2}$
$\partial F_y / \partial z$	$(m+2, n+1, o+2)$	$\mathbf{G}_z \otimes \mathbf{I}_{n+1} \otimes \mathbf{I}_{m+2}$
$\partial F_x / \partial z$	$(m+1, n+2, o+2)$	$\mathbf{G}_z \otimes \mathbf{I}_{n+2} \otimes \mathbf{I}_{m+1}$
$\partial F_z / \partial x$	$(m+2, n+2, o+1)$	$\mathbf{I}_{o+1} \otimes \mathbf{I}_{n+2} \otimes \mathbf{G}_x$
$\partial F_y / \partial x$	$(m+2, n+1, o+2)$	$\mathbf{I}_{o+2} \otimes \mathbf{I}_{n+1} \otimes \mathbf{G}_x$
$\partial F_x / \partial y$	$(m+1, n+2, o+2)$	$\mathbf{I}_{o+2} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+1}$

3.4 Assembled operator

Let $s_{F_x} = (m+1)(n+2)(o+2)$, $s_{F_y} = (m+2)(n+1)(o+2)$, $s_{F_z} = (m+2)(n+2)(o+1)$, and similarly s_{C_x} , s_{C_y} , s_{C_z} for the output edge sizes. The 3×3 block curl matrix acting on $\mathbf{F} = [F_x; F_y; F_z]$ is:

$$\mathbf{C}_{3D} = \begin{bmatrix} \mathbf{0}_{s_{C_x} \times s_{F_x}} & -(\mathbf{G}_z \otimes \mathbf{I}_{n+1} \otimes \mathbf{I}_{m+2}) & +(\mathbf{I}_{o+1} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+2}) \\ +(\mathbf{G}_z \otimes \mathbf{I}_{n+2} \otimes \mathbf{I}_{m+1}) & \mathbf{0}_{s_{C_y} \times s_{F_y}} & -(\mathbf{I}_{o+1} \otimes \mathbf{I}_{n+2} \otimes \mathbf{G}_x) \\ -(\mathbf{I}_{o+2} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+1}) & +(\mathbf{I}_{o+2} \otimes \mathbf{I}_{n+1} \otimes \mathbf{G}_x) & \mathbf{0}_{s_{C_z} \times s_{F_z}} \end{bmatrix}$$

Note the antisymmetric sign pattern: the diagonal blocks are zero, and off-diagonal blocks appear with opposite signs across the diagonal, mirroring the structure of the continuous curl as a cross product.

3.5 Curl of gradient identity in 3D

The full 3D gradient maps a scalar field ϕ at $(m+2)(n+2)(o+2)$ positions to face-staggered components:

$$\mathbf{G}_{\text{full}} = \begin{bmatrix} \mathbf{I}_{o+2} \otimes \mathbf{I}_{n+2} \otimes \mathbf{G}_x \\ \mathbf{I}_{o+2} \otimes \mathbf{G}_y \otimes \mathbf{I}_{m+2} \\ \mathbf{G}_z \otimes \mathbf{I}_{n+2} \otimes \mathbf{I}_{m+2} \end{bmatrix}.$$

By the same mixed-product argument as in 2D, each row of $\mathbf{C}_{3D} \mathbf{G}_{\text{full}}$ reduces to the difference of two identical Kronecker products and therefore vanishes:

$$\mathbf{C}_{3D} \mathbf{G}_{\text{full}} = \mathbf{0}.$$

4 Summary

Both operators are constructed entirely from the 1D mimetic gradient provided by MOLE, extended to multiple dimensions through Kronecker products with identity matrices. The key properties are:

1. The operators are **sparse** — the number of nonzeros per row is bounded by the stencil width of the 1D gradient.
2. The fundamental identity $\nabla \times (\nabla \phi) = 0$ holds **exactly** at the discrete level, by algebraic cancellation.
3. The operators inherit the **order of accuracy** k of the underlying mimetic gradient at interior nodes.